🏁 Final Review: Reading · ANSWER KEY

Biology A · Unit 8 · Level 3 (closer to the class work) · goes with End of semester: Final review + Final Exam

Name: _____

The Semester in Five Big Ideas

Science skills: goggles go on for chemicals, heat and glass, and every accident is reported. A hypothesis is a testable prediction (If… then… because…). The independent variable is what you change; the dependent variable is what you measure.

Orcas: the Southern Residents (J, K and L pods) eat Chinook salmon. PCBs biomagnify up the food web and are stored in blubber. Boat noise blocks echolocation, so hunting takes more energy.

Photosynthesis and respiration are partners. Photosynthesis (in chloroplasts): carbon dioxide + water + light → glucose + oxygen. Cellular respiration (in mitochondria): glucose + oxygen → carbon dioxide + water + ATP.

Energy and biofuel: algae store energy as oil (lipids), which can become biodiesel, a renewable fuel. Exothermic reactions release heat; endothermic reactions take in heat. The carbon cycle moves carbon among air, living things and fuels.

Food, diabetes and fossils: carbohydrates become glucose; insulin lets glucose into cells to keep blood sugar in balance (homeostasis). In type 1 diabetes the body makes little or no insulin; in type 2 cells do not respond well. Fossils form by fast burial in fine sediment, and deeper rock layers are older.

Check your reading

1. Compare the inputs and outputs of photosynthesis and respiration.
They are opposites: photosynthesis uses CO2 + water + light to make glucose + oxygen; respiration uses glucose + oxygen to make CO2 + water + ATP.

2. Why do orcas have the most PCBs?
Biomagnification: they are at the top of the food web.

3. What does insulin do?
Lets glucose into cells, lowering blood sugar.